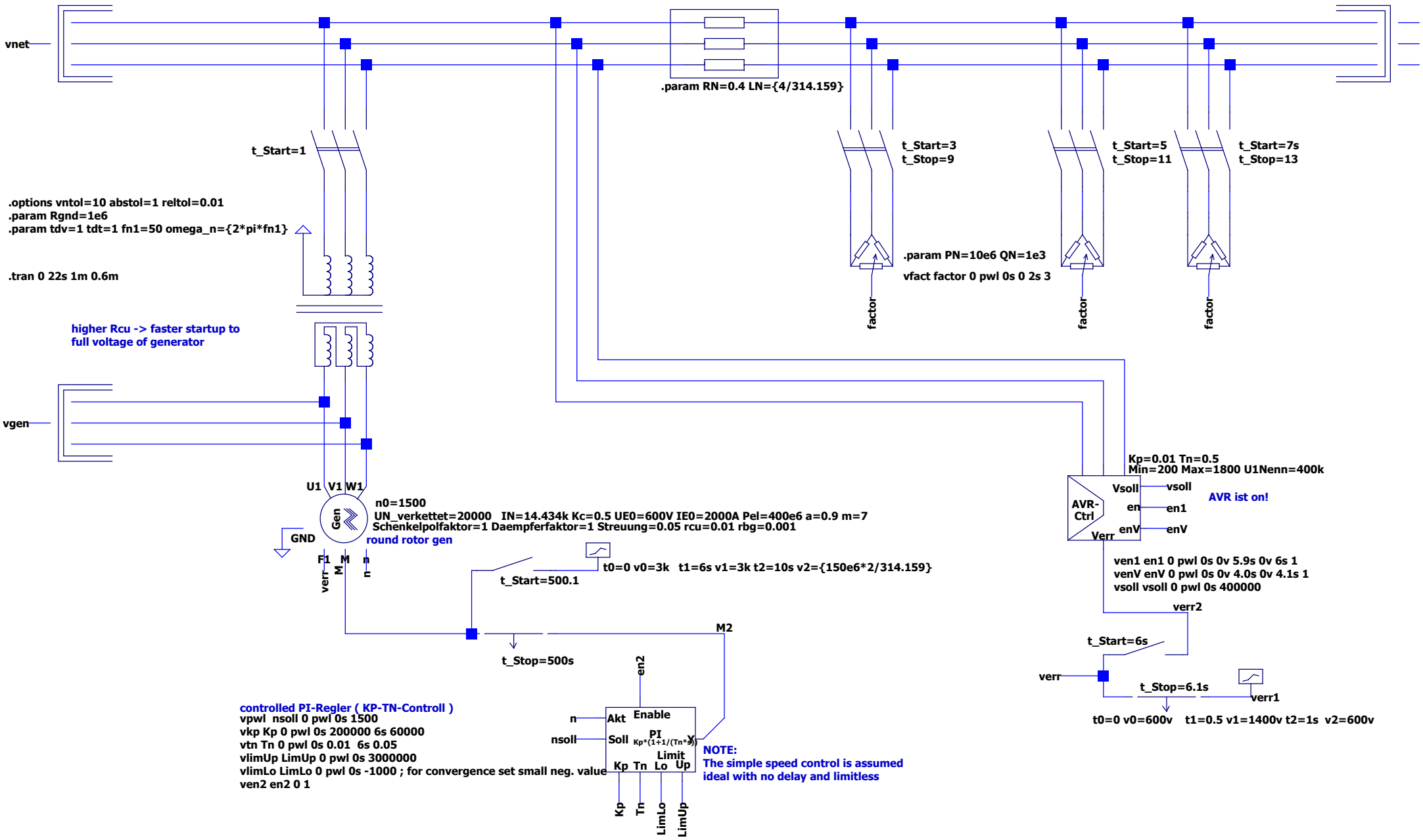
vnet
.param RN=0.4 LN={4/314.159}
t_Start=1
t_Start=3
t_Stop=9
t_Start=5
t_Stop=11
t_Start=7s
t_Stop=13
.options vntol=10 abstol=1 reltol=0.01
.param Rgnd=1e6
.param tdv=1 tdt=1 fn1=50 omega_n={2*pi*fn1}
.tran 0 22s 1m 0.6m
.param PN=10e6 QN=1e3
vfact factor 0 pwl 0s 0 2s 3
factor
factor
factor
higher Rcu -> faster startup to
full voltage of generator
vgen
U1 V1 W1
Gen
GND
n0=1500
UN_verkettet=20000 IN=14.434k Kc=0.5 UE0=600V IE0=2000A Pel=400e6 a=0.9 m=7
Schenkelpolfaktor=1 Daempferfaktor=1 Streuung=0.05 rcu=0.01 rbg=0.001
round rotor gen
F1 M n
verr M n
Kp=0.01 Tn=0.5
Min=200 Max=1800 U1Nenn=400k
AVR-
Ctrl
Vsoll
en
enV
Verr
vsoll
en1
enV
AVR ist on!
t0=0 v0=3k t1=6s v1=3k t2=10s v2={150e6*2/314.159}
t_Start=500.1
ven1 en1 0 pwl 0s 0v 5.9s 0v 6s 1
venV enV 0 pwl 0s 0v 4.0s 0v 4.1s 1
vsoll vsoll 0 pwl 0s 400000
verr2
M2
t_Start=6s
t_Stop=500s
verr
t_Stop=6.1s
verr1
t0=0 v0=600v t1=0.5 v1=1400v t2=1s v2=600v
en2
controlled PI-Regler (KP-TN-Controll)
vpwl nsoll 0 pwl 0s 1500
vkp Kp 0 pwl 0s 200000 6s 60000
vtn Tn 0 pwl 0s 0.01 6s 0.05
vlimUp LimUp 0 pwl 0s 3000000
vlimLo LimLo 0 pwl 0s -1000 ; for convergence set small neg. value
ven2 en2 0 1
n
nsoll
Akt
Soll
Enable
PI
Kp*(1+1/(Tn*s))
Limit
Kp Tn Lo Up
Y
NOTE:
The simple speed control is assumed
ideal with no delay and limitless
Kp
Tn
LimLo
LimUp